

Erkan Tairi

Education

- 2019–2024 **PhD in Computer Science**, *TU Wien*, Vienna, Austria
Supervisors: Matteo Maffei (TU Wien) and Daniel Slamanig (AIT Austrian Institute of Technology)
Thesis: Foundations of Adaptor Signatures for Distributed Ledger Protocols
<https://doi.org/10.34726/hss.2024.123264>
- 2016–2018 **MSc in Computer Science**, *Johannes Kepler University*, Linz, Austria
- 2011–2015 **BSc in Computer Science**, *University St. Paul the Apostle*, Ohrid, Macedonia

Professional Experience

- Oct. 2025 - **Postdoctoral Researcher**, *UC Berkeley*, Berkeley, CA, USA
Host: Sanjam Garg
- Sep. 2025 - **Research Scientist**, *Alloc Init*, Remote
Cryptographic protocol design and documentation
- May 2024 - **Postdoctoral Researcher**, *ENS Paris*, Paris, France
Sep. 2025 Host: David Pointcheval
- Feb. 2022 - **Research Engineer**, *MyPrivacy*, Vienna, Austria
Jan. 2024 Cryptographic protocol design and development

Internships and Visits

- May 2026 **Research Visit**, *ENS Paris*, Paris, France
Host: Brice Minaud
- Feb. 2025 **Research Visit**, *King's College London*, London, England
Host: Martin Albrecht
- Dec. 2022 **Research Visit**, *IMDEA Software Institute*, Madrid, Spain
Host: Dario Fiore
- Nov. 2022 **Research Visit**, *ETH Zürich*, Zürich, Switzerland
Host: Dennis Hofheinz
- Jun.-Sep. 2018 **Internship**, *AIT Austrian Institute of Technology*, Vienna, Austria
Supervisor: Daniel Slamanig
- Oct. 2018 - **Internship**, *TU Wien*, Vienna, Austria
Mar. 2019 Supervisor: Matteo Maffei

Teaching and Supervision

- 2019–2023 **Teaching Assistant**, *TU Wien*, Vienna, Austria
Introduction to Cryptography, Privacy-Enhancing Cryptography, Cryptocurrencies, (graduate level)
- Master Thesis Supervision**
Aron Wussler (TU Wien) – Post-quantum Cryptography in OpenPGP
<https://doi.org/10.34726/hss.2023.106226>
Ngaffo Raissa Flora (University of Bamenda) – Simulation-extractable NIZK Proof Systems

Publications

Published Papers

Lighthouse: Single-Server Secure Aggregation with $O(1)$ Server-Committee Communication. In USENIX Security 2026.

Sanjam Garg, Alireza Kavousi, Dimitris Kolonelos, [Erkan Tairi](#), and Zhipeng Wang

<https://eprint.iacr.org/2026/346>

LeOPaRd: Towards Practical Post-Quantum Oblivious PRFs via Interactive Lattice Problems. In CRYPTO 2026.

Muhammed F. Esgin, Ron Steinfeld, [Erkan Tairi](#) and Jie Xu

<https://eprint.iacr.org/2024/1615>

(Fine-Grained) Unbounded Inner-Product Functional Encryption from LWE. In SCN 2026.

Valerio Cini and [Erkan Tairi](#)

<https://eprint.iacr.org/2026/062>

Ciphertext-Updatable Attribute-based and Predicate Encryption from Lattices. In SCN 2026.

Robert Schädlich, Linda Scheu-Hachtel, [Erkan Tairi](#) and Yuejun Wang

Hardness of M-LWE with General Distributions and Applications to Leaky Variants. In PKC 2026.

Katharina Boudgoust, Corentin Jeudy, [Erkan Tairi](#) and Weiqiang Wen

<https://eprint.iacr.org/2025/1472>

On Verifiable Delay Functions from Time-lock Puzzles. In PKC 2026.

Hamza Abusalah, Karen Azari, Dario Fiore, Chethan Kamath and [Erkan Tairi](#)

<https://eprint.iacr.org/2025/1782>

Impossibility of VDFs in the ROM: The Complete Picture. In EUROCRYPT 2026.

Hamza Abusalah, Karen Azari, Chethan Kamath, [Erkan Tairi](#) and Maximilian von Consbruch

<https://eprint.iacr.org/2025/1773>

Lower Bounds for Lattice-based Compact Functional Encryption. In EUROCRYPT 2024.

[Erkan Tairi](#) and Akin Ünal

https://doi.org/10.1007/978-3-031-58723-8_9

(Inner-Product) Functional Encryption with Updatable Ciphertexts. In Journal of Cryptology.

Valerio Cini, Sebastian Ramacher, Daniel Slamanig, Christoph Striecks and [Erkan Tairi](#)

<https://doi.org/10.1007/s00145-023-09486-y>

Optimizing 0-RTT Key Exchange with Full Forward Security. In ACM CCSW 2023.

Christian Göth, Sebastian Ramacher, Daniel Slamanig, Christoph Striecks, [Erkan Tairi](#) and Alexander Zikulnig

<https://doi.org/10.1145/3605763.3625246>

LedgerLocks: A Security Framework for Blockchain Protocols Based on Adaptor Signatures. In ACM CCS 2023.

[Erkan Tairi](#), Pedro Moreno-Sanchez and Clara Schneidewind

<https://doi.org/10.1145/3576915.3623149>

Foundations of Coin Mixing Services. In ACM CCS 2022.

Noemi Glaeser, Matteo Maffei, Giulio Malavolta, Pedro Moreno-Sanchez, [Erkan Tairi](#) and Sri AravindaKrishnan Thyagarajan

<https://doi.org/10.1145/3548606.3560637>

A2L: Anonymous Atomic Locks for Scalability in Payment Channel Hubs. In IEEE S&P 2021.

[Erkan Tairi](#), Pedro Moreno-Sanchez and Matteo Maffei

<https://doi.org/10.1109/sp40001.2021.00111>

Post-Quantum Adaptor Signature for Privacy-Preserving Off-Chain Payments. In FC 2021.

[Erkan Tairi](#), Pedro Moreno-Sanchez and Matteo Maffei

https://doi.org/10.1007/978-3-662-64331-0_7

Updatable Signatures and Message Authentication Codes. In PKC 2021.
Valerio Cini, Sebastian Ramacher, Daniel Slamanig, Christoph Striecks and Erkan Tairi
https://doi.org/10.1007/978-3-030-75245-3_25

Preprints (Under Submission)

Registered Functional Encryption for Pseudorandom Functionalities from Lattices: Registered ABE for Unbounded Depth Circuits and Turing Machines, and More.

Tapas Pal, Robert Schädlich and Erkan Tairi

<https://eprint.iacr.org/2025/967>

Foundations of Verifiably Encrypted (Blind) Signatures.

Diego Castejon-Molina, Erkan Tairi, Dimitrios Vasilopoulos and Pedro Moreno-Sanchez

Adaptor Signatures Meet BLS: Efficient Blockchain Applications with Unique Adaptor Signatures.

Javier Gomez-Martinez, Erkan Tairi, Pedro Moreno-Sanchez and Clara Schneidewind

Professional Activities

Program Committee

ACM CCS 2025-2026, ACM AsiaCCS 2026, FC 2025-2026, IACR Communications in Cryptology 2025-2026, CVC 2025

External Reviewer

ACISP 2024; ACM CCS 2021-2023; ACM AFT 2022; ACNS 2024-2025; APKC 2021-2022; Asiacrypt 2021, 2024-2025; CANS 2022; Crypto 2023-2026; Eurocrypt 2023, 2025-2026; FC 2021-2024; IEEE S&P 2024; IWSEC 2021-2023; PKC 2026; ProvSec 2020-2023

Administration and Organization

- Co-organizer of ViSP Cryptography Research Meetup
- Member of NDSS 2023 Student Support Committee

Awards and Grants

Erwin Schrödinger Fellowship (by Austrian Science Fund)

<https://doi.org/10.55776/J4879>

France-Berkeley Fund

Input | Output Research Grant

(Selected) Presentations and Invited Talks

(Fine-Grained) Unbounded Inner-Product Functional Encryption from LWE

Crypto Day at Télécom Paris. Jun. 2025

<https://shorturl.at/DrJEr>

Guest talk at King's College London Cybersecurity Group. Feb. 2025

LedgerLocks: A Security Framework for Blockchain Protocols Based on Adaptor Signatures

IOG Seminar. Nov. 2023

(Inner-Product) Functional Encryption with Updatable Ciphertexts

PICOCRYPT Seminar at IMDEA Software Institute. Dec. 2022

<https://shorturl.at/5RP3o>

A2L: Anonymous Atomic Locks for Scalability in Payment Channel Hubs

IEEE Symposium on Security and Privacy. May 2021

<https://www.youtube.com/watch?v=RPj8efQtEyg>

Post-Quantum Adaptor Signature for Privacy-Preserving Off-Chain Payments

Financial Cryptography and Data Security 2021. Mar. 2021

<https://www.youtube.com/watch?v=BqMDnWFOGL0>

Decrypto Seminar. Dec. 2020

Updatable Signatures and Message Authentication Codes

Conference PKC 2021. May 2021

<https://www.youtube.com/watch?v=RqAlXixDZ1M>

Young Researcher Crypto Seminar. May 2021

Computer Skills

Programming	C, C++, Rust (<i>advanced</i>), C#, Python, Shell (<i>average</i>), Java, Go (<i>basics</i>)
Web	HTML, CSS, JavaScript (React, Node.js), ASP.NET Web (<i>advanced</i>)
Software	Visual Studio, SageMath, Magma, Git, $\text{\LaTeX}$
Other	Microsoft Azure, Amazon AWS (cloud computing)

Languages

Macedonian	Native	
English	Fluent	<i>professional working proficiency</i>
Turkish	Fluent	<i>professional working proficiency</i>
French	Intermediate	<i>elementary proficiency</i>
German	Intermediate	<i>elementary proficiency</i>
Albanian	Intermediate	<i>elementary proficiency</i>

Referees

Sanjam Garg	sanjamg@berkeley.edu
Daniel Slamanig	daniel.slamanig@unibw.de
David Pointcheval	david.pointcheval@ens.fr
Matteo Maffei	matteo.maffei@tuwien.ac.at